A very relevant property of square matrices relates to this: two real $n \times n$ matrices $A$ and $B$ are <u>similar</u>, denoted $A \sim B$, if there exists a real invertible $n \times n$ matrix $P$ such that

$$A = PBP^{-1}$$

Very relevant idea in higher level math in general, when dealing with a noncommutative operation, two elements $x$ and $y$ are <u>conjugates</u> if there exists element $z$ such that

$$x = zyz^{-1}$$

One big reason that can be relevant in general is the big reason we use it as well. That is, we wondered for $A$ any square matrix how can we compute $A^k$ ($k$ positive integer) in an efficient manner. Using the similar form:

$$A^k = (PBP^{-1})^k = (PBP^{-1})(PBP^{-1}) \cdot \ldots \cdot (PBP^{-1})(PBP^{-1})$$

So this gives $A^k = PB^k P^{-1}$

Say $B$ hypothetically is a much easier matrix to work with than $A$, that it's much easier to compute $B^k$ than $A^k$ e.g. if $B$ were a diagonal matrix that would be natural.

If so, the similarity between $A$ and $B$ would provide us a shortcut by which to compute $A^k$ in a more efficient manner.

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So similar matrices can clearly help us so it's useful to learn more about them. One very helpful thing is that two similar matrices have the same characteristic polynomial and therefore same eigenvalues (not necessarily same eigenvectors though). Why?

$$\begin{aligned} A - \lambda I &= PBP^{-1} - \lambda PP^{-1} \\ &= P(B - \lambda I)P^{-1} \end{aligned}$$

---> $$\det(A - \lambda I) = \det[P(B - \lambda I)P^{-1}]$$

$$= (\det P)[\det(B - \lambda I)](\det P^{-1})$$

since determinant of product is product of determinants.

Now can move terms around (real number multiplication is commutative, earlier we couldn't since matrix multiplication is not commutative). You had a HW question on this very thing, that's why they gave that, they were preluding to this exact issue.

So combining the two shown, this simplifies to

$$\det(A - \lambda I) = \det(B - \lambda I)$$

proving the result. So this gives a hint on how to find a matrix similar to A which is hopefully much simpler than A, that's what next section is about.

Before that: important to realize that two similar matrices do have the same characteristic polynomial / same eigenvalues, but two matrices with the same eigenvalues are NOT guaranteed to be similar (i.e. it is a necessary condition but not a sufficient one).

A good example (mentioned these two last time):

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad \text{vs.} \quad B = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

These two have the same eigenvalues as both are triangular ($A$ is diagonal in fact) with the same diagonal entries (which are the eigenvalues for a triangular matrix). But they are definitely not similar as $A$ being the identity matrix, it is similar to itself only since

$$P^{-1} I P = I$$

for any invertible $P$.

# 5.3 Diagonalization

Real $n \times n$ matrix $A$ is <u>diagonalizable</u> (over $\mathbb{R}$) if there exists real $n \times n$ matrices $P$ and $D$, $P$ invertible, $D$ diagonal such that

$$A = PDP^{-1}$$

That is, $A$ is similar to the diagonal matrix $D$.
If possible, this is quite helpful, already alluded to one way ($k$ is any positive integer) this can help:

$$A^k = PD^k P^{-1}$$

easy to compute

That is, if you want the high power of a generic square matrix, diagonalizing it (assuming that's possible) would be a convenient way to find that high power, hopefully without doing a ton of calculation.

So for any real $n \times n$ matrix $A$ two natural questions:

(1) Is $A$ diagonalizable over $\mathbb{R}$

(2) If $A$ is diagonalizable, how do we find its diagonalization in an efficient manner.

Fortunately answering first question also largely answers the second. Such $A$ is diagonalizable if and only if it has $n$ linearly independent real eigenvectors. Why? Say

$$A = PDP^{-1}$$

Then that means $AP = PD$. Suppose

$$P = \underbrace{\begin{bmatrix} v_1 & v_2 & \cdots & v_n \end{bmatrix}}_{\text{columns}} \quad , \quad D = \begin{bmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{bmatrix}$$

Via matrix multiplication, columns of $AP$ are:

$$Av_1, Av_2, \ldots, Av_n$$

while columns of $PD$ are:

$$\lambda_1 v_1, \lambda_2 v_2, \ldots, \lambda_n v_n$$

So for these two matrices to equal each other we must have:

$$\begin{aligned} Av_1 &= \lambda_1 v_1 \\ Av_2 &= \lambda_2 v_2 \\ &\vdots \\ Av_n &= \lambda_n v_n \end{aligned}$$

Since none of these $v_j'$ s is 0 (they are columns of $P$ which is stated to be invertible), this by definition means $\lambda_1, \lambda_2, \ldots, \lambda_n$ are eigenvalues of $A$ with $v_1, v_2, \ldots, v_n$ associated eigenvectors for them.

So this proves that if $A$ is diagonalizable, it must have $n$ linearly independent eigenvectors (they must be linearly independent to make $P$ invertible as required).

This also shows how to create the diagonalization when those $n$ linearly independent eigenvectors exist.

That is, take $P$ to have columns those $n$ eigenvectors and $D$ with diagonal entries the associated eigenvalues of the corresponding columns in $P$.

So $D$ is unique up to order of entries, $P$ as well up to order of columns, though can also multiply any column in $P$ by a nonzero constant since it stays an eigenvector.

So in short, to diagonalize, find all eigenvalues and their associated eigenvectors. Assuming enough of the latter, that creates $P$ and the former creates $D$.

Remember that a set of eigenvectors, where each eigenvector corresponds to a different eigenvalue is automatically a linearly independent set (we proved that earlier, and this is main reason we want that).

So unless some of the eigenvectors correspond to the same eigenvalue, we should be safe in regards to linear independence.

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5.3.11

$$A = \begin{bmatrix} -1 & 4 & -2 \\ -3 & 4 & 0 \\ -3 & 1 & 3 \end{bmatrix} \quad , \quad \lambda = 1, 2, 3$$

$$A - I = \begin{bmatrix} -2 & 4 & -2 \\ -3 & 3 & 0 \\ -3 & 1 & 2 \end{bmatrix} \text{----->} \begin{bmatrix} -1 & 2 & -1 \\ -1 & 1 & 0 \\ -3 & 1 & 2 \end{bmatrix}$$

via scaling top two rows. Then row equivalent to

$$\begin{bmatrix} -1 & 2 & -1 \\ 0 & -1 & 1 \\ 0 & -5 & 5 \end{bmatrix} \text{---->} \begin{bmatrix} -1 & 2 & -1 \\ 0 & -1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

In first step via replacing latter two rows with themselves plus appropriate multiple of first. Then replace third row by itself minus five times second.

This now being in REF, has two pivots so null space (eigenspace) will have dimension 1 so just find one nonzero vector in it and we are set. Equations are

$$\begin{aligned} -x_1 + 2x_2 - x_3 &= 0 \\ -x_2 + x_3 &= 0 \end{aligned}$$

Since only need one vector, plug in a simple nonzero value for one of them and find rest from there.

If we were needing to find multiple (independent) solutions it might be wiser to proceed more systematically and find the parametric vector form, but since we need only one here, brute forcing it is fairly efficient.

$$x_3 = 1 \text{ ----> } x_2 = 1 \text{ ----> } x_1 = 1$$

Hence $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ is an eigenvector for $\lambda = 1$

Next we have

$$A - 2I = \begin{bmatrix} -3 & 4 & -2 \\ -3 & 2 & 0 \\ -3 & 1 & 1 \end{bmatrix} \text{ ----> } \begin{bmatrix} -3 & 4 & -2 \\ 0 & -2 & 2 \\ 0 & -3 & 3 \end{bmatrix}$$

via replacing latter two rows by themselves minus the first. Now scale second and third to make leading 1 in each. Then replace third by itself minus second to get

$$\begin{bmatrix} -3 & 4 & -2 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$

Again dimension 1 for null space so find single eigenvector in same way:

$$\begin{aligned} -3x_1 + 4x_2 - 2x_3 &= 0 \\ x_2 - x_3 &= 0 \end{aligned}$$

Again plug in simple nonzero value for one of them:

$$x_3 = 1 \text{ ----> } x_2 = 1 \text{ ----> } x_1 = \frac{2}{3}$$

Hence $\begin{bmatrix} 2 \\ 3 \\ 3 \end{bmatrix}$ is an eigenvector for $\lambda = 2$

(multiply by 3 just to clean it up).

And then for the final eigenvalue:

$$A - 3I = \begin{bmatrix} -4 & 4 & -2 \\ -3 & 1 & 0 \\ -3 & 1 & 0 \end{bmatrix} \text{----->} \begin{bmatrix} -2 & 2 & -1 \\ -3 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

via replacing third row by itself minus second and scaling first. Again null space dimension 1 so

$$\begin{aligned} -2x_1 + 2x_2 - x_3 &= 0 \\ -3x_1 + x_2 &= 0 \end{aligned}$$

which yields:

$$x_1 = 1 \text{ ----> } x_2 = 3 \text{ ----> } x_3 = 4$$

Thus $\begin{bmatrix} 1 \\ 3 \\ 4 \end{bmatrix}$ is an eigenvector for $\lambda = 3$

Won't find $P^{-1}$ since that's not really a new idea in this.

From these:

$$P = \begin{bmatrix} 1 & 2 & 1 \\ 1 & 3 & 3 \\ 1 & 3 & 4 \end{bmatrix} \quad , \quad D = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

Note that you can put the columns of $P$ in any order as long as you change the columns in $D$ accordingly, the diagonal entry in a particular column of $D$ should have an eigenvector corresponding to it in that same column in $P$.

So there's no really "correct" order in that respect, you can do what you want, though certainly it's more natural to do it in some type of logical order, i.e. say in increasing or decreasing order.

These questions can be quite long! If eigenvalues are not given have to find those on top of all this. Plus for the full diagonalization need $P^{-1}$ as well.

Diagonalizing a 3 x 3 matrix is the most common setting for this class, but for a generic matrix, it can be pretty involved! Realistically have to make sure the eigenvalues can be found via reasonable methods as you can't be expected to solve a generic cubic.

But even if it is designed as such, still work to find those, and then have to find eigenvector(s) for each.

Once you have those, you have $P$ and $D$, but then in theory have to find $P^{-1}$ as well which is no cake walk for 3 x 3.

So oftentimes I would only ask you to find $P$ and $D$ as that's what's truly needed to prove you know how to diagonalize a matrix as again the questions can just get too long if you have to do them in full.

But if question asked to compute say $A^{100}$ then you'd have no choice but to find $P^{-1}$, so depends on what the problem is asking for.

So does this idea always work to diagonalize any real, square matrix? Say for a 3 x 3 we can always find the three eigenvalues and their corresponding eigenvectors, same idea as last question? Not so fast!

5.3.13

$$A = \begin{bmatrix} 2 & 2 & -1 \\ 1 & 3 & -1 \\ 1 & -2 & 2 \end{bmatrix}, \qquad \lambda = 1, 5$$

Only two eigenvalues here? But why? Seems like characteristic polynomial is of degree three so should have three roots still?

What if one root is repeated? Probably the case here!

This makes things trickier. We found one eigenvector for each eigenvalues in last problem. If we just do that here will only have two eigenvectors when we need three?

Does that mean not possible to diagonalize here? Not so simple! What if one eigenvalue has two linearly independent eigenvectors, i.e. its eigenspace is of dimension higher than 1. Well when is that even possible? Sure didn't happen in any of the prior problems that we did.

So this suggests two potential issues that could come into play here:

(1) Having a "repeated" eigenvalue (presumably a real one).
(2) Nonreal eigenvalues

This semi suggests that if our eigenvalues are distinct real numbers then we ought to be safe, but ok that too should be clarified completely. But mostly want to assess how the situation is when you have a repeated real eigenvalue and how to deal with it properly.

Big issue is that we are checking on the nullity of a matrix, i.e. how many linearly independent eigenvectors we can obtain from a specific eigenvalue. But as you've seen, nullity can vary a great deal from one matrix to another so this will require careful examination to understand.